

Subgroup coverage in split conformal classification: marginal validity, Mondrian repair, and small-group failure, with an audit of TabPFN

Nadim Mahmud Dipu^a

^a*Brac University, Dhaka, Bangladesh*

Abstract

Split conformal prediction endows any classifier with finite-sample, distribution-free *marginal* coverage, but marginal validity says nothing about how miscoverage is distributed across protected subgroups—the property that matters when prediction sets gate consequential decisions. We study this gap, empirically and through a finite-sample variance argument, on three US Census tasks from folktabs (ACSIIncome, ACSEmployment, ACSPublicCoverage; California), wrapping five predictors in split conformal prediction across three confidence levels, two conformity scores, and four group taxonomies. Marginal coverage conceals systematic subgroup miscoverage: at a 90% target the worst-covered age group receives only 86.2% coverage (gap 0.081), though every model meets the marginal guarantee. Mondrian (group-conditional) calibration repairs this where strata are large—the age gap falls to 0.051 ($p < 10^{-10}$), costing 0.021 in mean set size—but *worsens* it where strata are small: on the race axis, whose smallest calibration stratum holds only ≈ 84 points, the gap rises from 0.046 to 0.057 ($p < 10^{-3}$). We trace this to the finite-sample variance of the in-expectation guarantee: realized per-group coverage fluctuates as $\sqrt{\alpha(1-\alpha)/n_g}$, so the max–min gap grows

as strata shrink. These effects are nearly identical across all five predictors—coverage fairness is governed by the conformal procedure and the group structure, not the model. As the model under audit, TabPFN is the most reliable base learner we tested: better calibrated out of the box than untuned GBDTs (adaptive ECE 0.026 vs. 0.075/0.052), needing no temperature scaling, and yielding the smallest sets. We also flag a silent failure of non-randomized APS on binary tasks, and give support-aware guidance: use LAC, report worst-group coverage, and apply Mondrian only where $n_g \gtrsim 10/\alpha$.

Keywords: conformal prediction, equalized coverage, group-conditional calibration, Mondrian conformal prediction, uncertainty quantification, calibration, algorithmic fairness, tabular foundation models, TabPFN

1. Introduction

Set-valued prediction with coverage guarantees has become a standard tool for trustworthy machine learning. Given any classifier and an exchangeable calibration sample, split (inductive) conformal prediction returns a prediction set that contains the true label with a user-specified probability $1 - \alpha$, distribution-free and in finite samples [1, 2, 3]. This guarantee, however, is *marginal*: it averages over the whole population. In the high-stakes tabular settings where conformal prediction is increasingly deployed—credit scoring, benefits eligibility, clinical triage—decisions concern individuals who belong to protected demographic groups, and a guarantee that holds on average can fail conditionally. A 90% marginal guarantee is compatible with, say, 86% coverage for one group and 94% for another, silently concentrating error—less reliable, less informative sets—on the worse-covered group [4]. Whether,

and when, marginal conformal prediction hides such subgroup disparities is
15 the central question of this paper.

The textbook remedy is *Mondrian* (group-conditional) conformal prediction [5, 4], which calibrates a separate quantile within each group and so guarantees coverage *within* every group—“equalized coverage”—rather than only on average. That guarantee, though, holds in expectation over calibration
20 tion draws, and it is purchased with data: each group’s threshold is estimated from only the n_g calibration points falling in that group. When some groups are small—as protected attributes with imbalanced categories routinely are—those thresholds are noisy, and the realized per-group coverage can vary enough that group-conditional calibration *increases* the empirical
25 disparity it was meant to remove. The practically important question is therefore not whether equalized coverage holds in expectation (it does), but when Mondrian stratification helps versus harms in finite samples, and how a practitioner should decide. We give a clean empirical characterization of both regimes together with a simple finite-sample variance argument that
30 predicts which one obtains.

We situate this study in the tabular regime and pair it with a second, timely question: which base learner to trust. For over a decade gradient-boosted decision trees (GBDTs) have been the default for tabular classification [6], but they are now challenged by *tabular foundation models*, most
35 prominently TabPFN, a prior-data fitted network (PFN) that performs supervised learning as a single forward pass of in-context inference and reportedly matches or exceeds tuned GBDT ensembles on small-to-medium tables [7, 8]. The choice of base learner still matters for uncertainty: although

the conformal quantile step repairs *validity* for any model, the calibration
 40 of the underlying probabilities controls set *efficiency*, and good probabilities
 are what make calibrated decisions and tight sets possible. TabPFN’s
 posterior-predictive outputs are often *described* as well calibrated by construction,
 yet we are aware of no matched-protocol audit of (a) how that calibration compares to strong
 baselines, (b) how its probabilities behave as
 45 conformity scores inside split conformal prediction, and (c) whether the coverage
 it induces is fair across protected groups. We therefore adopt TabPFN as the model under
 audit—a strong, modern base learner whose uncertainty we scrutinize alongside the
 conformal questions above, against XGBoost, LightGBM, and an MLP. For GBDTs and
 conventional neural networks the
 50 calibration literature is mature [9, 10]; for tabular foundation models it is nascent.

We study three binary prediction tasks derived from US Census micro-
 data via the folktables package [11]—ACSIIncome, ACSEmployment, and
 ACSPublicCoverage for California—each carrying protected attributes (sex,
 55 race, age). We compare TabPFN, with and without post-hoc temperature scaling,
 against XGBoost, LightGBM, and an MLP (all with fixed, untuned default
 hyperparameters) under an identical train/calibration/test protocol with five
 random seeds, and we wrap every model in split conformal prediction at three
 target levels (80%, 90%, 95%) using two conformity scores (LAC
 60 and APS), in both marginal and Mondrian (group-conditional) variants.

Our research questions and headline answers are:

- **RQ1 (Calibration).** *Are TabPFN’s native probabilities better calibrated than those of untuned GBDTs and an MLP?* Yes, substantially

so relative to GBDTs: pooled adaptive ECE is 0.026 for TabPFN ver-
65 sus 0.075 (XGBoost) and 0.052 (LightGBM), with paired Wilcoxon
 $p < 10^{-4}$; TabPFN is statistically indistinguishable from the MLP
($p = 0.60$) while being more accurate, and temperature scaling adds
no significant improvement ($p = 0.12$).

- **RQ2 (Conformal validity and efficiency).** *Do all models achieve*
70 *nominal marginal coverage, and which yields the most efficient (small-*
est) prediction sets? With the LAC score, empirical coverage tracks all
three targets to within one percentage point for every model; TabPFN
yields the smallest sets at every level (e.g., 1.296 vs. 1.327–1.363 at
90%; $p < 10^{-3}$ against each baseline). We additionally document that
75 the APS score *without randomization* is degenerate on these binary
tasks, returning the full label set almost surely—a practically impor-
tant negative result.

- **RQ3 (Coverage fairness).** *Does marginal conformal prediction hide*
subgroup miscoverage, and does Mondrian conformal repair it? Marginal
80 coverage conceals an age-related gap of 0.081 at the 90% target (worst-
group coverage 0.862). Mondrian stratification repairs this for the
age axis (gap $\rightarrow$ 0.051, worst-group coverage $\rightarrow$ 0.881, $p < 10^{-10}$)
at a small efficiency cost (+0.021 mean set size), is neutral on the
already-equitable sex axis, but *increases* the empirical gap on the race
85 axis (from 0.046 to 0.057, $p < 10^{-3}$) without improving worst-group
coverage—a finite-sample failure mode driven by small calibration strata.
Crucially, these effects are nearly identical across all five models: cov-

erage fairness is governed by the conformal procedure and the group structure, not by the choice of predictor.

90 Our contributions are: (1) an empirical demonstration, across five predictors and three census tasks, that marginal conformal validity conceals systematic subgroup miscoverage, together with a finite-sample variance characterization of when group-conditional (Mondrian) calibration repairs it—large strata—versus *worsens* it—small strata—including a documented, statistically significant small-group failure on the race axis, and the finding
95 that these effects are essentially model-independent; (2) a practically important negative result on non-randomized APS for binary tasks; (3) the first matched-protocol audit, to our knowledge, of a tabular foundation model’s calibration and conformal efficiency against GBDT and MLP baselines (in
100 their standard untuned configurations), establishing that TabPFN is better calibrated than GBDTs—needing no temperature scaling—and yields the smallest prediction sets; (4) concrete, support-aware deployment guidance (use LAC, report worst-group coverage, and apply Mondrian only where $n_g \gtrsim 10/\alpha$); and (5) an open, fully scripted pipeline with committed per-seed
105 result files and a verification script that recomputes every reported number, enabling exact reproduction.

The remainder of the paper is organized as follows. Section 2 reviews related work and positions our novelty. Section 3 formalizes the calibration metrics, split conformal prediction, the LAC and APS scores, and Mondrian
110 conformal prediction. Section 4 details datasets, models, protocol, and the statistical analysis plan. Section 5 reports results for RQ1–RQ3. Section 6 interprets the findings and derives deployment guidance, Section 7 discusses

threats to validity, and Section 8 concludes.

2. Related work

115 *Conformal prediction..* Split (inductive) conformal prediction converts any scoring model into set-valued predictions with finite-sample, distribution-free marginal coverage under exchangeability [1, 2, 3]. For classification, the LAC/homogeneous score thresholds $1 - \hat{p}_y$ [12], while APS accumulates sorted class probabilities to improve conditional coverage [13]; APS requires
120 tie-breaking randomization to avoid over-coverage, a detail our results show to be decisive on binary tasks.

Group-conditional coverage and fairness in uncertainty quantification.. Equalized coverage asks that prediction sets cover each protected group at the nominal rate, achievable by Mondrian/group-conditional calibration [4, 5].
125 Subsequent work studies class-conditional and group-conditional validity, its finite-sample costs, and approximate conditional coverage [14, 15, 16]. The guarantee is an expectation over calibration draws, and its finite-sample slack and variance grow as strata shrink; yet empirical audits of *coverage disparity* have targeted conventional models, none (to our knowledge) includes a
130 tabular foundation model, and the regime in which Mondrian calibration *backfires* on small strata has received little direct empirical study—a gap our RQ3 addresses head-on.

Calibration of machine-learning classifiers.. Modern neural networks are frequently miscalibrated and benefit from post-hoc recalibration such as temperature scaling [9]; boosted trees are known to push scores away from the
135

extremes, motivating Platt scaling and isotonic regression [10]. Metric estimation itself is delicate: binned ECE is biased, and adaptive (equal-mass) binning is the recommended estimator [17]. We adopt adaptive ECE as our primary metric and report Brier score and NLL as proper-scoring comple-
140 ments.

Tabular foundation models and prior-data fitted networks.. PFNs meta-train a transformer on synthetic datasets drawn from a prior over data-generating mechanisms, so that supervised prediction on a new table reduces to in-context inference [18]. TabPFN instantiated this idea for small tabular clas-
145 sification [7] and its successor scaled it to larger tables and regression [8]. Evaluations of these models have focused overwhelmingly on accuracy and AUC against GBDT baselines [6, 19]; their uncertainty behaviour is usually asserted (via the Bayesian-posterior interpretation) rather than audited.

Benchmarking tabular deep learning under matched protocols.. A recurring
150 lesson of tabular benchmarking is that conclusions are protocol-sensitive—tuning budgets, preprocessing, and split design can reverse rankings [6, 20]. We therefore hold the entire pipeline fixed across models and vary only the predictor.

Novelty.. The intersection of these strands is empty: existing TabPFN evalu-
155 ations do not audit calibration or conformal behaviour against matched baselines; the conformal-fairness literature has not examined foundation-model conformity scores; and tabular benchmarks do not report group-conditional coverage. We fill precisely this gap, in the regime where TabPFN is intended to operate (binary/multiclass tasks of moderate size with demographic struc-

160 ture), and we report the failure modes (APS degeneracy, Mondrian small-group noise) alongside the successes.

3. Background and preliminaries

We consider classification with covariates $X \in \mathcal{X}$, labels $Y \in \mathcal{Y} = \{1, \dots, K\}$ (here $K = 2$), and, for fairness analyses, a protected attribute
 165 $A \in \mathcal{A}$. A probabilistic classifier outputs $\hat{p}(y \mid x)$.

3.1. Calibration metrics

A classifier is *top-label calibrated* if $\Pr(Y = \hat{y} \mid \hat{p}(\hat{y} \mid X) = c) = c$ for all confidence levels c , where $\hat{y} = \arg \max_y \hat{p}(y \mid x)$. The expected calibration error partitions test predictions into B bins $\{\mathcal{B}_b\}$ by confidence and computes

$$\text{ECE} = \sum_{b=1}^B \frac{|\mathcal{B}_b|}{n} \left| \text{acc}(\mathcal{B}_b) - \text{conf}(\mathcal{B}_b) \right|, \quad (1)$$

170 where acc and conf are the mean accuracy and mean confidence within a bin. We use *adaptive* (equal-mass) binning with $B = 15$ bins as the primary estimator and report equal-width ECE in the released CSVs; we additionally report the maximum calibration error (MCE), the Brier score $\frac{1}{n} \sum_i \|\hat{p}(\cdot \mid x_i) - e_{y_i}\|_2^2$, and the negative log-likelihood (NLL), both strictly proper scoring
 175 rules. *Temperature scaling* [9] post-processes logits z as $\text{softmax}(z/T)$ with a single scalar $T > 0$ fit on the calibration split by NLL minimization; it is monotone and therefore accuracy-preserving.

3.2. Split conformal prediction

Split conformal prediction [2, 1] uses a held-out calibration set $\{(x_i, y_i)\}_{i=1}^n$,
 180 disjoint from training data, and a conformity score $s(x, y)$ measuring how

poorly y conforms at x (larger = worse). Let

$$\hat{q} = \text{the } \frac{\lceil (n+1)(1-\alpha) \rceil}{n}\text{-th empirical quantile of } \{s(x_i, y_i)\}_{i=1}^n. \quad (2)$$

The prediction set for a test point x is $C(x) = \{y : s(x, y) \leq \hat{q}\}$. If the calibration and test points are exchangeable, then for any score function and any model,

$$1 - \alpha \leq \Pr(Y_{\text{test}} \in C(X_{\text{test}})) \leq 1 - \alpha + \frac{1}{n+1}, \quad (3)$$

185 where the upper bound holds when scores are almost surely distinct [21, 3]. Coverage is *marginal*: it averages over the population and need not hold conditionally on subgroups.

3.3. Conformity scores: LAC and APS

We evaluate two standard classification scores. *LAC* (least ambiguous
190 set-valued classifier) [12]sets

$$s_{\text{LAC}}(x, y) = 1 - \hat{p}(y \mid x), \quad (4)$$

which yields the smallest average set size among procedures with marginal validity when probabilities are correct, but offers no conditional-coverage adaptivity. *APS* (adaptive prediction sets) [13]sorts classes by predicted probability, $\hat{p}_{(1)} \geq \dots \geq \hat{p}_{(K)}$, and scores

$$s_{\text{APS}}(x, y) = \sum_{j: \hat{p}_{(j)} > \hat{p}(y|x)} \hat{p}_{(j)} + u \cdot \hat{p}(y \mid x), \quad u \sim \text{Unif}(0, 1), \quad (5)$$

195 where the randomization term u is essential for exact validity; the deterministic variant ($u \equiv 1$, i.e., including the full mass of the label's own class) is

conservative. Our implementation uses the deterministic variant (the score accumulates sorted class probabilities up to and including the true label’s own mass, with no tie-breaking randomization), and Section 5.2 shows this is not a cosmetic detail on binary tasks.

3.4. *Mondrian (group-conditional) conformal prediction and equalized coverage*

Given a partition of the population by a taxonomy $g(x, a) \in \mathcal{G}$ (here: the protected attribute, or the predicted class), Mondrian conformal prediction [5] computes the quantile in Eq. (2) *separately within each group*, using the n_g calibration points of group g . This yields the group-conditional guarantee

$$\Pr(Y \in C(X) \mid g(X, A) = g) \geq 1 - \alpha \quad \text{for every } g \in \mathcal{G}, \quad (6)$$

i.e., *equalized coverage* in expectation [4]. The price is statistical, and it is the crux of our RQ3. Each group’s threshold $\hat{q}_g$ is estimated from only $n_g \ll n$ calibration points; and because the realized per-group coverage on the test set is an average of (approximately) Bernoulli($1 - \alpha$) indicators, its standard deviation about the nominal level is

$$\text{sd}(\hat{c}_g) \approx \sqrt{\frac{\alpha(1-\alpha)}{n_g}}. \quad (7)$$

The empirical max–min coverage gap over $|\mathcal{G}|$ groups is therefore dominated by the smallest stratum, growing like $1/\sqrt{\min_g n_g}$, and the guarantee’s finite-sample slack $1/(n_g + 1)$ likewise grows as groups shrink. Equalized coverage *in expectation* (Eq. (6)) is thus fully compatible with a *larger realized gap* than marginal calibration produces whenever strata are small—the effect our RQ3 isolates and Eq. (7) quantifies.

3.5. Fairness metrics for prediction sets

For groups $g \in \mathcal{G}$ with empirical per-group coverage c_g and mean set
220 size $\bar{k}_g$ on the test set, we report the *coverage gap* $\max_g c_g - \min_g c_g$, the
worst-group coverage $\min_g c_g$, and the *set-size disparity* $\max_g \bar{k}_g - \min_g \bar{k}_g$.

4. Methodology

4.1. Datasets and protected attributes

We use three binary prediction tasks from the folktables suite [11], all
225 on California ACS microdata (2018 1-Year person-survey release): **ACSIncome** (predict income $>$ \$50k), **ACSEmployment** (predict employment status), and **ACSPublicCoverage** (predict public health-insurance coverage). These tasks were introduced precisely as fairness-aware replacements for UCI Adult and carry the protected attributes we audit: *sex* (2 groups,
230 derived from the binary SEX field), *race* (from RAC1P, coarsened to four groups—White, Black, Asian, and a merged *Other* stratum collecting all remaining RAC1P codes), and *age* (AGEP binned into five bands with edges at 25, 40, 55, and 65: <25 , 25–39, 40–54, 55–64, 65+). ACSPublicCoverage is defined by folktables only for respondents under 65, so it carries four age
235 bands rather than five. In addition to the three protected axes we evaluate a *predicted-class* taxonomy (Mondrian on $\hat{y}$), a standard label-conditional stratification. Per-group calibration-set sizes are reported in Table 1; they are the key to the race-axis result in Section 5.3.

4.2. Preprocessing

240 Numeric features are median-imputed and categorical features ordinal-encoded (unseen categories mapped to -1), yielding a single `float32` design

Table 1: Per-group calibration-set sizes n_g (seed 0; $n_{\text{cal}} = 2,000$ per dataset, summing each block to that total). Counts shift by at most a few percent across the five seeds. The race axis contains a far smaller minority stratum (Black, $n_g \approx 84\text{--}131$) than any age band, which drives the Mondrian failure of Section 5.3. ACSPublicCoverage has no 65+ band by task definition.

Axis	Group	ACSEmployment	ACSIIncome	ACSPublicCoverage
Sex	Female	1046	961	1112
	Male	954	1039	888
Race	White	1271	1194	1122
	Asian	312	342	308
	Other	333	378	439
	Black	84	86	131
Age	<25	577	231	651
	25–39	393	664	557
	40–54	406	625	450
	55–64	272	340	342
	65+	352	140	—

matrix; only the MLP additionally standardizes features to zero mean and unit variance inside its own pipeline. Each task is randomly subsampled, with the seed, to $n = 8,000$ rows to remain within TabPFN’s supported
245 small-data regime. All five models receive identical feature matrices within a cell. We note explicitly that the protected attributes are *not* withheld: SEX, RAC1P, and AGE1P belong to each folktasks task’s standard feature set and are therefore supplied to every model as ordinary inputs; we use them *additionally* for stratified evaluation and Mondrian calibration. Coverage

250 disparities thus arise even though every model can see the protected attribute,
which is the realistic setting for these administrative tasks.

4.3. Models

We compare five predictors: **TabPFN** (tabpfn package v8.0.7, default classifier checkpoint `tabpfn-v3-classifier-v3_default.ckpt`; default settings, no per-dataset tuning, in keeping with its intended usage); **TabPFN+T**,
255 the same posterior probabilities post-processed by temperature scaling fit on the calibration split; **XGBoost** (300 trees, max depth 6, learning rate 0.1, row/column subsampling 0.9, histogram tree method) and **LightGBM** (300 trees, 31 leaves, learning rate 0.05, row/column subsampling 0.9); and
260 an **MLP** (two hidden layers of 128 and 64 units, standardized inputs, early stopping on a 10% internal validation split, at most 300 epochs). All baseline hyperparameters are fixed at these common default values; no per-dataset hyperparameter search was performed for any model, so the comparison is between out-of-the-box configurations rather than tuned ones. Because tem-
265 perature scaling is a monotone transform of the probabilities, TabPFN and TabPFN+T induce (near-)identical conformal prediction sets; their conformal rows differ only through quantile tie effects (maximum observed difference in any coverage-gap cell: 0.0027). We therefore discuss them jointly in RQ2–RQ3 and separately only for calibration (RQ1).

270 4.4. Train / calibration / test protocol

For each (dataset, seed) cell we draw a label-stratified three-way split into proper training (45%), calibration (25%), and test (30%) sets—3,600, 2,000, and 2,400 rows respectively of the 8,000 sampled per cell (the committed

results record $n_{\text{test}} = 2,400$, confirming these proportions). The model is
 275 fit on the training split only; temperature and all conformal quantiles are
 computed on the calibration split only; every reported metric is computed
 on the untouched test split. Splits are redrawn with seeds $\{0, \dots, 4\}$, and all
 five models share the identical splits within a cell, enabling paired statistical
 tests.

280 4.5. Experiment matrix

The full matrix is: 3 datasets $\times$ 5 models $\times$ 5 seeds $\times$ 3 miscoverage levels
 $\alpha \in \{0.05, 0.10, 0.20\}$ (targets 95%/90%/80%) $\times$ 2 conformity scores (LAC,
 APS) $\times$ 4 group taxonomies (sex, race, age, predicted class) $\times$ 2 conformal
 methods (marginal, Mondrian), yielding 3,600 conformal evaluation rows and
 285 75 calibration rows, all committed to the repository.

4.6. Metrics

Calibration (RQ1): accuracy, adaptive ECE (primary), equal-width ECE,
 MCE, Brier, NLL. *Validity/efficiency (RQ2)*: empirical marginal coverage
 and mean prediction-set size. *Fairness (RQ3)*: coverage gap, worst-group
 290 coverage, and set-size disparity as defined in Section 3, for marginal vs. Mon-
 drian calibration on each protected axis.

4.7. Statistical protocol

Unless stated otherwise, point estimates are means over the relevant cells
 and dispersions are standard deviations across the 15 (dataset $\times$ seed) cells.
 295 Pairwise model comparisons use two-sided Wilcoxon signed-rank tests paired
 at the (dataset $\times$ seed) level ($n = 15$; the smallest attainable two-sided p is

Table 2: RQ1 — Accuracy and calibration, mean $\pm$ std over 15 (dataset $\times$ seed) cells.

Lower is better for all metrics except accuracy. Best in **bold**.

Model	Accuracy $\uparrow$	ECE _{ada} $\downarrow$	MCE $\downarrow$	Brier $\downarrow$	NLL $\downarrow$
XGBoost	0.763 ± 0.052	0.075 ± 0.019	0.135 ± 0.022	0.333 ± 0.064	0.516 ± 0.084
LightGBM	0.769 ± 0.049	0.052 ± 0.012	0.100 ± 0.018	0.317 ± 0.057	0.485 ± 0.074
MLP	0.759 ± 0.046	0.025 ± 0.007	0.057 ± 0.022	0.324 ± 0.056	0.486 ± 0.074
TabPFN	0.782 ± 0.046	0.026 ± 0.006	0.063 ± 0.021	0.298 ± 0.056	0.452 ± 0.075
TabPFN+T	0.782 ± 0.046	0.025 ± 0.006	0.054 ± 0.018	0.297 ± 0.056	0.451 ± 0.074

6.1×10^{-5} , which we report as $p < 10^{-4}$). Marginal-vs.-Mondrian comparisons are paired at the (dataset $\times$ model $\times$ seed) level ($n = 75$). Omnibus differences across the five models use the Friedman test over the 15 cells with
300 average ranks reported; given only five Tier-A model columns we report ranks descriptively rather than a full Nemenyi critical-difference analysis, which would require the larger Tier-B suite that is not part of the committed run (see Section 7). We treat $p < 0.01$ as significant in view of the multiplicity of comparisons.

305 5. Results

5.1. RQ1: TabPFN is natively well calibrated; GBDTs are not

Table 2 reports pooled calibration metrics; Table 3 breaks adaptive ECE out by dataset; Fig. 1 shows a representative reliability diagram.

Three observations follow. *First*, both GBDTs are substantially miscal-
310 ibrated out of the box, with the familiar pattern of over-confidence at high confidence levels (Fig. 1): XGBoost’s pooled adaptive ECE is 0.075 and

Table 3: RQ1 — Adaptive ECE by dataset (mean $\pm$ std over 5 seeds).

Model	ACSEmployment	ACSIIncome	ACSPublicCoverage
XGBoost	0.058 ± 0.005	0.069 ± 0.006	0.098 ± 0.009
LightGBM	0.043 ± 0.006	0.049 ± 0.007	0.064 ± 0.010
MLP	0.020 ± 0.006	0.024 ± 0.003	0.033 ± 0.007
TabPFN	0.022 ± 0.005	0.025 ± 0.005	0.030 ± 0.007
TabPFN+T	0.021 ± 0.004	0.025 ± 0.004	0.028 ± 0.007

LightGBM’s 0.052, against 0.026 for TabPFN. The paired differences are significant at the resolution limit of the test (TabPFN vs. XGBoost and vs. LightGBM: Wilcoxon $p < 10^{-4}$ in both cases, with TabPFN better in all 15
315 of 15 cells). *Second*, TabPFN’s native calibration matches a well-regularized MLP (median paired ECE difference -0.001 , $p = 0.60$) while being more accurate (paired accuracy difference $+0.023$, $p < 10^{-3}$) and strictly better on both proper scoring rules (Brier and NLL, $p < 10^{-4}$ against all three base-
lines). The Friedman omnibus test on adaptive ECE is highly significant
320 ($\chi^2 = 48.5$, $p = 7.5 \times 10^{-10}$) with average ranks TabPFN+T 1.8, MLP 2.0, TabPFN 2.2, LightGBM 4.0, XGBoost 5.0. *Third*, post-hoc temperature scaling improves TabPFN only marginally (pooled ECE $0.026 \rightarrow 0.025$; paired
 $p = 0.12$): within the datasets studied, TabPFN does not need recalibration, which is consistent with—and, to our knowledge, the first matched-baseline
325 statistical evidence for—the posterior-predictive interpretation of PFN outputs.

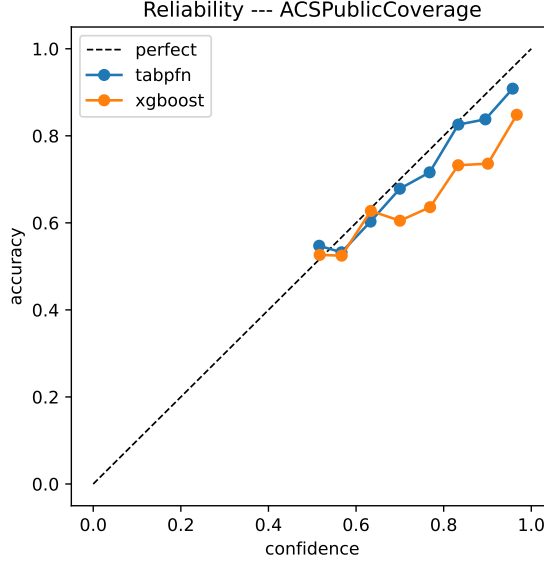


Figure 1: Reliability diagram on ACSPublicCoverage-CA (seed 0). XGBoost (orange) sits below the diagonal at high confidence—systematic over-confidence—while TabPFN (blue) tracks the diagonal closely. Curves use 15 equal-width confidence bins; the dashed line is perfect calibration.

5.2. *RQ2: All models are conformally valid; TabPFN is the most efficient.*

Non-randomized APS is degenerate on binary tasks

Table 4 reports marginal coverage and mean set size under the LAC
 330 score. Empirical coverage tracks the nominal target for every model and
 level, as Eq. (3) guarantees: pooled over models, coverage is 0.805 ± 0.013 ,
 0.901 ± 0.009 , and 0.951 ± 0.007 at the 80%, 90%, and 95% targets respectively.
 Conformal validity is thus model-agnostic, as theory predicts—miscalibrated
 GBDT probabilities are repaired by the quantile step.

335 Efficiency, by contrast, is model-dependent and rewards probability qual-
 ity. TabPFN produces the smallest prediction sets at every target level; at

Table 4: RQ2 — Marginal coverage and mean set size under the LAC score (mean over 15 cells per row). Coverage within one point of target for all models; smaller sets are better. Best set size in **bold**. TabPFN+T is omitted: temperature scaling is monotone, so its conformal sets coincide with TabPFN’s.

Model	Target 80%		Target 90%		Target 95%	
	Cov.	Size	Cov.	Size	Cov.	Size
XGBoost	0.800	1.083	0.899	1.339	0.952	1.534
LightGBM	0.802	1.072	0.901	1.327	0.953	1.519
MLP	0.803	1.095	0.903	1.363	0.951	1.543
TabPFN	0.809	1.058	0.900	1.296	0.950	1.480

the 90% level its mean set size of 1.296 improves on LightGBM (1.327), XGBoost (1.339), and the MLP (1.363), with paired Wilcoxon $p = 4.3 \times 10^{-4}$, 3.1×10^{-4} , and 6.5×10^{-4} respectively. On a binary task, set size minus one
340 is the singleton-failure rate, so this difference is operationally meaningful: at the 90% level TabPFN returns an uninformative two-label set on 29.6% of cases versus 36.3% for the MLP—a 6.7-point reduction in abstentions at identical coverage.

A cautionary negative result on APS. Under our deterministic APS imple-
345 mentation, every model at every level returns essentially the full label set: pooled coverage is 1.000 and mean set size is 1.99–2.00 at all three targets. The mechanism is elementary but easy to overlook: on a binary task the deterministic APS score of the true label is the cumulative probability mass *including* the label’s own class, which equals 1 whenever the true label is the
350 lower-ranked class; the calibration quantile at any usable α therefore satu-

rates at (or near) 1, and the resulting sets contain both labels. Randomized APS [13] avoids this by interpolating the final class’s mass. Practitioners wrapping binary tabular classifiers should either use LAC or ensure their APS implementation randomizes; we flag this because deterministic APS is the default in some libraries and the failure is silent—coverage looks excellent while the sets are vacuous. All fairness analyses below therefore use the LAC score; the corresponding APS rows are retained in the released CSVs but are uninformative by construction.

5.3. RQ3: Marginal coverage hides subgroup gaps; Mondrian repairs large groups but harms small ones

Table 5 is the core fairness result: per-axis coverage gap, worst-group coverage, and mean set size at the 90% target under LAC, pooled over the five models and 15 cells, for marginal versus Mondrian calibration. Table 6 shows the coverage gap across all three target levels. Figs. 2–4 display the per-model breakdown.

Marginal validity is not subgroup validity.. Although every model satisfies the marginal 90% guarantee, per-group coverage is far from uniform. On the age axis the pooled coverage gap is 0.081 and worst-group coverage drops to 0.862—i.e., the least-covered age group experiences roughly 38% more mis-coverage than nominal. At the 80% target the age gap widens to 0.151 with worst-group coverage 0.732. The race axis shows a smaller but persistent gap (0.046 at 90%, worst-group coverage 0.875), while the sex axis is essentially equitable already (0.012). These disparities are a property of the population/score interaction, not of the predictor: per-model gaps under marginal

Table 5: RQ3 — Subgroup coverage fairness at the 90% target (LAC; pooled over 5 models $\times$ 3 datasets $\times$ 5 seeds; marginal vs. Mondrian paired tests with $n = 75$). Gap = max–min per-group coverage; WGC = worst-group coverage. Arrows mark the better direction; p -values are two-sided Wilcoxon for the marginal-vs.-Mondrian contrast.

Axis	Method	Gap $\downarrow$	WGC $\uparrow$	Mean size $\downarrow$	
Sex	Marginal	0.012	0.895	1.324	gap $p = 0.50$ WGC $p = 0.79$
	Mondrian	0.013	0.895	1.327	
Age	Marginal	0.081	0.862	1.324	gap $p = 4.2 \times 10^{-11}$ WGC $p = 2.9 \times 10^{-10}$
	Mondrian	0.051	0.881	1.345	
Race	Marginal	0.046	0.875	1.324	gap $p = 9.3 \times 10^{-4}$ WGC $p = 0.55$
	Mondrian	0.057	0.873	1.334	
Pred. class	Marginal	0.045	0.881	1.324	
	Mondrian	0.013	0.894	1.332	

375 calibration at 90% on age lie in the narrow band 0.078–0.083 across all five models, TabPFN included. A foundation model’s better probabilities buy smaller sets (RQ2) but do not, by themselves, buy equitable coverage.

Mondrian repairs the age axis at a small price.. Group-conditional calibration on age reduces the pooled gap from 0.081 to 0.051 and raises worst-group 380 coverage from 0.862 to 0.881 (both $p < 10^{-9}$, $n = 75$), at a mean set-size cost of +0.021 ($1.324 \rightarrow 1.345$, $p < 10^{-12}$)—about two additional both-label sets per hundred predictions. The same pattern holds at the 80% target (gap $0.151 \rightarrow 0.106$; worst-group coverage $0.732 \rightarrow 0.774$). The predicted-class taxonomy behaves similarly (gap $0.045 \rightarrow 0.013$). For TabPFN specifically, 385 Mondrian-on-age lifts worst-group coverage from 0.862 to 0.881 while its sets

Table 6: RQ3 — Coverage gap by target level (LAC; pooled over models, datasets, seeds). Mondrian helps consistently on age, is neutral-to-mixed on sex, and does not help—and at 90% significantly hurts—on race.

	Sex		Age		Race	
Target	Marg.	Mond.	Marg.	Mond.	Marg.	Mond.
80%	0.029	0.021	0.151	0.106	0.056	0.055
90%	0.012	0.013	0.081	0.051	0.046	0.057
95%	0.012	0.010	0.045	0.037	0.034	0.038

remain smaller than any baseline’s marginal sets—the efficiency advantage survives the fairness repair.

Mondrian fails on the small-support race axis.. On race, Mondrian *increases* the empirical gap at the 90% target from 0.046 to 0.057 ($p = 9.3 \times 10^{-4}$) and
390 leaves worst-group coverage unchanged (0.875 vs. 0.873, $p = 0.55$); the same direction holds, non-significantly, at 95%. The effect is consistent across datasets (Employment 0.034 $\rightarrow$ 0.049; Income 0.047 $\rightarrow$ 0.060; PublicCover-
age 0.056 $\rightarrow$ 0.061) and across all five models (Fig. 4). The mechanism is the finite-sample cost identified in Section 3: the race taxonomy contains groups
395 with small calibration support—the Black stratum holds only $n_g \approx 84$ –131 calibration points across the three tasks, against 1,100–1,300 for White (Table 1)—so each small group’s threshold $\hat{q}_g$ is a high quantile estimated from few points. The group-conditional guarantee of Eq. (6) still holds *in expectation over calibration draws*, but the realized per-group coverage fluctuates
400 with standard deviation $\approx \sqrt{\alpha(1-\alpha)/n_g}$, and the max–min gap statistic grows with the number of groups and with $1/\sqrt{\min_g n_g}$. Equalized coverage

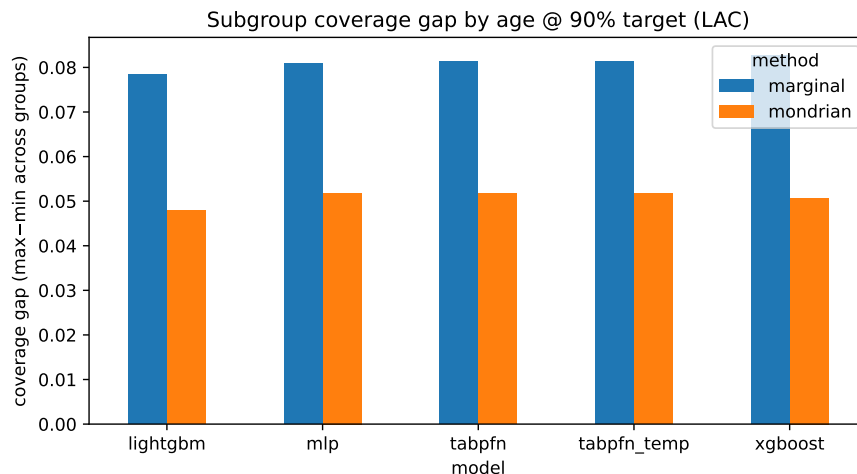


Figure 2: Coverage gap by age at the 90% target, marginal (blue) vs. Mondrian (orange), per model (LAC score). Mondrian reduces the gap by roughly 0.03 uniformly across all five models, TabPFN included.

in expectation is therefore compatible with—and here empirically produces—*larger* realized disparities than marginal calibration when strata are small. Sex (two large groups) shows neither benefit nor harm because marginal coverage is already nearly equalized.

6. Discussion

What the audit establishes about TabPFN.. Within the studied regime—binary census tasks, moderate sample sizes, five seeds—TabPFN’s uncertainty is trustworthy on the first two criteria of Section 1: its native probabilities are as calibrated as anything we tested without any post-processing, and they translate into the most efficient conformal sets. Both findings are consistent with the PFN design: a model meta-trained to approximate a posterior predictive should produce probabilities whose *ordering and sharp-*

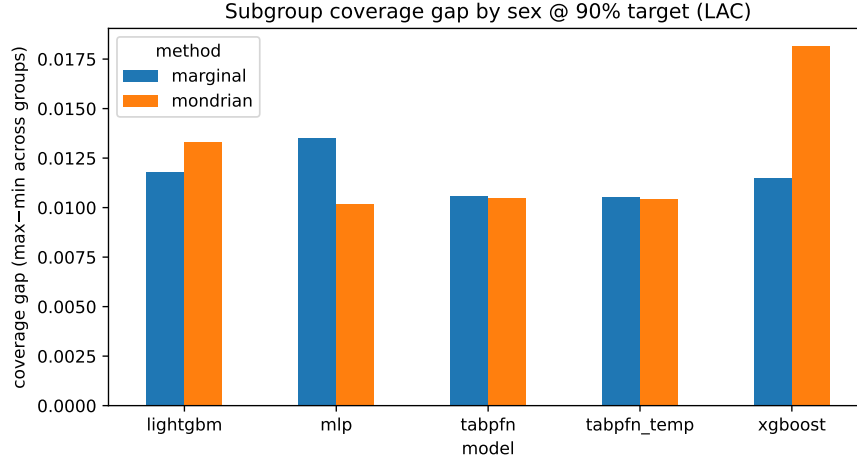


Figure 3: Coverage gap by sex at the 90% target (LAC score). Gaps are an order of magnitude smaller than on age and statistically indistinguishable between methods.

ness reflect genuine predictive uncertainty, and LAC efficiency is exactly a
 415 function of probability quality. The practical upshot is that the usual GBDT
 deployment recipe—train, then Platt/isotonic/temperature recalibrate—can
 be dropped for TabPFN in this regime; temperature scaling moved pooled
 ECE by one point in the third decimal and was statistically indistinguishable
 from the identity.

420 *What it establishes about conformal fairness..* The third criterion—equitable
 uncertainty—is not a model property at all in our data. Coverage gaps under
 marginal calibration are nearly identical across the five predictors, and so is
 the effect of Mondrian stratification. The decision that matters is procedural:
which taxonomy to condition on, given its calibration support. Our results
 425 give an unusually clean empirical demonstration of both sides: conditioning
 on age (few, large groups, with a real coverage disparity to fix) halves the gap

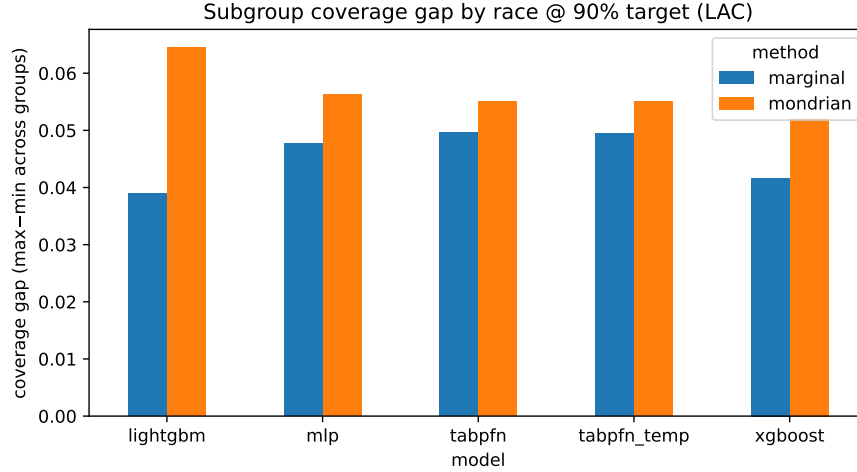


Figure 4: Coverage gap by race at the 90% target (LAC score). Mondrian (orange) *exceeds* marginal (blue) for every model: with small calibration strata, group-conditional thresholds are noisy and realized disparities grow.

at the cost of 0.02 extra labels per set; conditioning on race (many groups, some very small) backfires, inflating realized disparity without improving the worst group. We emphasize that this is not a contradiction of the equalized-coverage guarantee, which is an expectation over calibration randomness; it is the finite-sample variance of that guarantee made visible. Reporting only the expectation-level guarantee, as is common, would have concealed it.

Deployment guidance.. For practitioners deploying small-to-medium tabular classifiers with demographic structure, our results support the following sequence. (1) If TabPFN’s scale limits permit, prefer it: best accuracy, best proper scores, no recalibration step. (2) Wrap whatever model is chosen in split conformal prediction with the *LAC* score; verify that any APS implementation randomizes before using it on low-cardinality label spaces. (3)

Audit per-group coverage on a held-out set before choosing between marginal
 440 and Mondrian calibration: if a gap exists *and* every group has adequate calibration support, use Mondrian on that axis. As a rule of thumb, a group needs enough points for its $1 - \alpha$ quantile to be stable—on the order of $n_g \gtrsim 10/\alpha$ (about 100 at $\alpha = 0.1$). Our axes straddle this line precisely: the smallest age band (65+, $n_g = 140$ in ACSIncome; Table 1) clears it
 445 and Mondrian helps, whereas the smallest race stratum (Black, $n_g = 84$ in ACSEmployment) falls below it and Mondrian backfires. If some groups are small, prefer marginal calibration, merge small strata, or adopt methods designed for approximate conditional validity [15, 16]. (4) Report worst-group coverage alongside marginal coverage; the latter alone certified all our models
 450 while one age group received 73% coverage at an 80% target.

TabPFN versus GBDTs, honestly.. Our comparison favours TabPFN on every uncertainty metric, but three caveats temper a blanket recommendation. The tasks sit squarely inside TabPFN’s comfort zone (3,600 training rows per cell, numeric/categorical census features, binary labels); the GBDTs re-
 455 ceived *no* hyperparameter tuning—they ran with the fixed default settings of Section 4, and a tuning budget (e.g., NLL-based early stopping or a learning-rate/depth search) would likely close part of the calibration gap; and GBDT inference is orders of magnitude cheaper at scale. Where these constraints bind, the correct reading of our results is narrower but still useful: GBDT
 460 probabilities should not be consumed raw, and conformal wrapping equalizes validity (though not efficiency) across model families.

7. Limitations and threats to validity

External validity. All results derive from three binary tasks on Californian ACS microdata; conclusions are scoped to this regime and should not be
465 extrapolated to multiclass problems, other states or survey years, non-census domains, or tables beyond TabPFN’s supported size. The planned Tier-B evaluation over a broader OpenML suite, including Friedman–Nemenyi ranking across heterogeneous datasets, was not part of the committed results and remains future work. *Construct validity.* ECE estimates depend on binning;
470 we mitigate with adaptive binning and proper scoring rules, which agree in direction. The max–min coverage gap is sensitive to the number and size of groups, which is precisely the phenomenon RQ3 documents; worst-group coverage is the more decision-relevant statistic. *Internal validity.* GBDT and MLP hyperparameters were fixed at common default values with no
475 per-dataset tuning; a larger tuning budget could narrow the calibration and efficiency gaps. Five seeds bound our resolution; all headline effects are nonetheless significant at $p < 10^{-3}$ under paired tests. Temperature scaling and conformal quantiles share the same calibration split, a standard but not innocuous choice. *Statistical validity.* Wilcoxon tests at $n = 15$ have a floor
480 of $p = 6.1 \times 10^{-5}$; Friedman ranks over five models and 15 cells are descriptive. We did not run distribution-shift experiments (e.g., cross-state transfer), so exchangeability holds by construction here and coverage under shift is untested. *Ethical scope.* Equalizing coverage is one narrow fairness criterion; it neither implies nor substitutes for fairness of the underlying decisions, and
485 the race categories in ACS data are coarse administrative constructs.

8. Conclusion and future work

We showed that marginal conformal validity is not subgroup validity. Across five predictors and three census tasks, split conformal prediction met its marginal coverage guarantee while concealing an eight-point age cover-
age gap common to all models; group-conditional (Mondrian) calibration
490 repaired this where strata were large but *backfired* on the small-support race axis—a concrete, statistically significant demonstration that equalized coverage in expectation can coexist with larger realized disparities. These effects were governed by the conformal procedure and the group structure rather than the model, and a simple finite-sample variance argument ($\sqrt{\alpha(1-\alpha)/n_g}$)
495 predicts which regime obtains. We also documented a silent failure of non-randomized APS on binary tasks. As the timely model under audit, TabPFN proved the most reliable base learner we tested: its native probabilities were as calibrated as a well-regularized MLP and far better than untuned GBDTs, required no temperature scaling, and yielded the smallest conformal predic-
500 tion sets at every coverage level. Together these results give the first matched-protocol picture of a tabular foundation model’s uncertainty *and* a clean characterization of when group-conditional conformal calibration should be trusted.

505 Future work proceeds on three fronts: (i) the Tier-B OpenML extension with Friedman–Nemenyi ranking to test generality across heterogeneous tabular domains; (ii) distribution-shift stress tests (cross-state and temporal transfer within ACS) probing coverage degradation when exchangeability fails, together with adaptive remedies [22, 23]; and (iii) a clinical-tabular
510 extension, where TabPFN’s small-data strengths and the cost asymmetry of

miscoverage make equitable uncertainty both harder and more consequential, and where small protected strata will make the Mondrian failure mode we identified the central methodological obstacle.

Reproducibility statement

515 All code, notebooks (smoke test, experiment runner, analysis), and the complete per-seed result files (`calibration.csv`, 75 rows; `conformal_fairness.csv`, 3,600 rows) are released at <https://github.com/Nadim-Mahmud-23/tabpfn-uncertainty-audit> under the MIT license, with the following pinned dependency versions: `tabpfn` 8.0.7 (PyTorch 2.9.1 backend, classifier checkpoint `tabpfn-v3-classifier-v3_default.ckpt`),
520 `xgboost` 3.2.0, `lightgbm` 4.6.0, `scikit-learn` 1.8.0, `folktables` 0.0.12, `numpy` 2.3.5, and `scipy` 1.17.0 under Python 3.11+. Every table and statistic in this paper is regenerable from the committed CSVs by the analysis scripts; figures regenerate from the same files. Data are downloaded automatically from the US Census Bureau via `folktables`; no manual data placement is
525 required. All experiments run on a single Apple Silicon MacBook with no GPU (CPU/MPS backend); the full experiment matrix completes in approximately one hour of wall-clock time (about 55 minutes for the 75-cell run, dominated by TabPFN’s CPU inference), well within the reach of commodity hardware.

530 Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

535 **Nadim Mahmud Dipu:** Conceptualization, Methodology, Software,
Validation, Formal analysis, Investigation, Data curation, Writing – original
draft, Writing – review & editing, Visualization.

Data availability

The data used in this study are publicly available from the U.S. Census
540 Bureau American Community Survey (ACS) Public Use Microdata Sample
(PUMS), 2018 1-Year release, accessed programmatically via the folktables
package [11]. All source code, notebooks, and the committed per-seed result
files needed to reproduce every table and figure are released in the repository
cited in the Reproducibility statement.

545 References

- [1] V. Vovk, A. Gammerman, G. Shafer, Algorithmic Learning in a Random
World, Springer, New York, 2005.
- [2] H. Papadopoulos, K. Proedrou, V. Vovk, A. Gammerman, Inductive
confidence machines for regression, in: Machine Learning: ECML 2002,
550 13th European Conference on Machine Learning, Vol. 2430 of Lecture
Notes in Computer Science, Springer, 2002, pp. 345–356. doi:10.1007/
3-540-36755-1_29.
- [3] A. N. Angelopoulos, S. Bates, Conformal prediction: A gentle intro-
duction, Foundations and Trends in Machine Learning 16 (4) (2023)
555 494–591. doi:10.1561/2200000101.

- [4] Y. Romano, R. F. Barber, C. Sabatti, E. J. Candès, With malice toward none: Assessing uncertainty via equalized coverage, *Harvard Data Science Review* 2 (2) (2020). doi:10.1162/99608f92.03f00592.
- [5] V. Vovk, D. Lindsay, I. Nouretdinov, A. Gammernan, Mondrian confidence machine, On-line Compression Modelling Project, Working Paper 4, Royal Holloway, University of London (Mar. 2003).
 URL <https://pure.royalholloway.ac.uk/en/publications/mondrian-confidence-machine>
- [6] L. Grinsztajn, E. Oyallon, G. Varoquaux, Why do tree-based models still outperform deep learning on typical tabular data?, in: *Advances in Neural Information Processing Systems 35 (NeurIPS 2022) Datasets and Benchmarks Track*, 2022.
 URL http://papers.nips.cc/paper_files/paper/2022/hash/0378c7692da36807bdec87ab043cdadc-Abstract-Datasets_and_Benchmarks.html
- [7] N. Hollmann, S. Müller, K. Eggenberger, F. Hutter, TabPFN: A transformer that solves small tabular classification problems in a second, in: *The Eleventh International Conference on Learning Representations (ICLR)*, 2023. arXiv:2207.01848.
 URL https://openreview.net/forum?id=cp5PvcI6w8_
- [8] N. Hollmann, S. Müller, L. Purucker, A. Krishnakumar, M. Körfer, S. B. Hoo, R. T. Schirrmeister, F. Hutter, Accurate predictions on small data with a tabular foundation model, *Nature* 637 (8045) (2025) 319–326. doi:10.1038/s41586-024-08328-6.

- 580 [9] C. Guo, G. Pleiss, Y. Sun, K. Q. Weinberger, On calibration of modern neural networks, in: Proceedings of the 34th International Conference on Machine Learning (ICML), Vol. 70 of Proceedings of Machine Learning Research, PMLR, 2017, pp. 1321–1330.
URL <https://proceedings.mlr.press/v70/guo17a.html>
- 585 [10] A. Niculescu-Mizil, R. Caruana, Predicting good probabilities with supervised learning, in: Proceedings of the 22nd International Conference on Machine Learning (ICML), ACM, 2005, pp. 625–632. doi:10.1145/1102351.1102430.
- [11] F. Ding, M. Hardt, J. Miller, L. Schmidt, Retiring adult: New datasets for fair machine learning, in: Advances in Neural Information Processing Systems 34 (NeurIPS 2021), 2021, pp. 6478–6490. arXiv:2108.04884.
590 URL <https://dblp.org/rec/conf/nips/DingHMS21.html>
- [12] M. Sadinle, J. Lei, L. Wasserman, Least ambiguous set-valued classifiers with bounded error levels, Journal of the American Statistical Association 114 (525) (2019) 223–234. doi:10.1080/01621459.2017.1395341.
595
- [13] Y. Romano, M. Sesia, E. J. Candès, Classification with valid and adaptive coverage, in: Advances in Neural Information Processing Systems 33 (NeurIPS 2020), 2020. arXiv:2006.02544.
URL <https://proceedings.neurips.cc/paper/2020/hash/244edd7e85dc81602b7615cd705545f5-Abstract.html>
600
- [14] T. Ding, A. N. Angelopoulos, S. Bates, M. I. Jordan, R. J. Tibshirani, Class-conditional conformal prediction with many classes, in: Advances

- in Neural Information Processing Systems 36 (NeurIPS 2023), 2023. `arXiv:2306.09335`.
- 605 URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/cb931eddd563f8d473c355518ce8601c-Abstract-Conference.html
- [15] I. Gibbs, J. J. Cherian, E. J. Candès, Conformal prediction with conditional guarantees, arXiv preprint (2023). `arXiv:2305.12616`.
- 610 URL <https://arxiv.org/abs/2305.12616>
- [16] C. Jung, G. Noarov, R. Ramalingam, A. Roth, Batch multivalid conformal prediction, in: The Eleventh International Conference on Learning Representations (ICLR), 2023. `arXiv:2209.15145`.
- URL <https://openreview.net/forum?id=Dk7QQp8jHEo>
- 615 [17] J. Nixon, M. W. Dusenberry, L. Zhang, G. Jerfel, D. Tran, Measuring calibration in deep learning, in: IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPR Workshops), 2019, pp. 38–41.
- URL https://openaccess.thecvf.com/content_CVPRW_2019/html/Uncertainty_and_Robustness_in_Deep_Visual_Learning/Nixon_Measuring_Calibration_in_Deep_Learning_CVPRW_2019_paper.html
- 620 [18] S. Müller, N. Hollmann, S. P. Arango, J. Grabocka, F. Hutter, Transformers can do bayesian inference, in: The Tenth International Conference on Learning Representations (ICLR), 2022. `arXiv:2112.10510`.
- 625 URL <https://openreview.net/forum?id=KSugKcbNf9>

- [19] D. C. McElfresh, S. Khandagale, J. Valverde, V. P. C., G. Ramakrishnan, M. Goldblum, C. White, When do neural nets outperform boosted trees on tabular data?, in: Advances in Neural Information Processing Systems 36 (NeurIPS 2023) Datasets and Benchmarks Track, 2023.
- 630 URL http://papers.nips.cc/paper_files/paper/2023/hash/f06d5ebd4ff40b40dd97e30cee632123-Abstract-Datasets_and_Benchmarks.html
- [20] Y. Gorishniy, I. Rubachev, V. Khrulkov, A. Babenko, Revisiting deep learning models for tabular data, in: Advances in Neural Information Processing Systems 34 (NeurIPS 2021), 2021. arXiv:2106.11959.
- 635 URL <https://proceedings.neurips.cc/paper/2021/hash/9d86d83f925f2149e9edb0ac3b49229c-Abstract.html>
- [21] J. Lei, M. G’Sell, A. Rinaldo, R. J. Tibshirani, L. Wasserman, Distribution-free predictive inference for regression, Journal of the American Statistical Association 113 (523) (2018) 1094–1111. doi: 10.1080/01621459.2017.1307116.
- 640
- [22] I. Gibbs, E. J. Candès, Adaptive conformal inference under distribution shift, in: Advances in Neural Information Processing Systems 34 (NeurIPS 2021), 2021, pp. 1660–1672. arXiv:2106.00170.
- 645 URL https://proceedings.neurips.cc/paper_files/paper/2021/hash/0d441de75945e5acbc865406fc9a2559-Abstract.html
- [23] R. J. Tibshirani, R. F. Barber, E. J. Candès, A. Ramdas, Conformal prediction under covariate shift, in: Advances in Neural Information Processing Systems 32 (NeurIPS 2019), 2019, pp. 2526–2536.
- 650

arXiv:1904.06019.

URL [https://proceedings.neurips.cc/paper/2019/hash/
8fb21ee7a2207526da55a679f0332de2-Abstract.html](https://proceedings.neurips.cc/paper/2019/hash/8fb21ee7a2207526da55a679f0332de2-Abstract.html)